

Filip BLAŠKOVIĆ



PROFILE

PhD researcher in Mathematics at Freie Universität Berlin and the Zuse Institute Berlin (ZIB), currently specializing in mathematical modeling, complex dynamical systems, network science, and computational methods. Experienced in analyzing complex datasets, designing efficient algorithms, and implementing scientific software in Python. Additional experience in developing predictive models for financial time-series forecasting, as well as building and maintaining containerized applications and data infrastructure using Docker and Kubernetes. Interested in quantitative research, data science, quantitative finance, forecasting, and algorithmic problem-solving roles.

CONTACT DETAILS

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PERSONAL INFORMATION

Citizenship:
Hungary (EU), Serbia
Languages:
English (C2),
German (B2.2),
Hungarian (B1),
Serbian (native)

PROFESSIONAL EXPERIENCE

RESEARCHER at Zuse Institute Berlin (Germany)

2023.08–pres.

- ◇ Develops mathematical and computational models for analyzing large-scale time-evolving network data.
- ◇ Designs algorithms for detecting structural patterns and changes in complex dynamical systems.
- ◇ Implements simulations and data-analysis pipelines in Python.
- ◇ Conducts independent research resulting in peer-reviewed publications and international conference presentations.
- ◇ *DAKI-FWS Early-warning systems project (2023)*: Developed and maintained containerized data infrastructure using Docker and Kubernetes, integrating Invenio, LakeFS, and Apache Superset for data management, versioning, and analytics workflows.

DATA ANALYST INTERN at Sinfon (Belgrade, Serbia)

2023.04–2023.08

- ◇ Developed predictive models for forecasting expenditure components of the national budget of the Republic of Serbia.
- ◇ Analyzed historical financial and economic time-series data.
- ◇ Performed model fitting, validation, and forecasting performance assessment.

TEACHING ASSISTANT at University of Novi Sad (Serbia)

2022.03–2023.08

- ◇ Conducted research in discrete mathematics and combinatorics.
- ◇ Communicated technical concepts through teaching, presentations, and mentoring activities.

EDUCATION

PHD IN MATHEMATICS. Freie Universität Berlin, Germany.

2023–present

- ◇ Research areas: mathematical modeling, network science, dynamical systems, and data analysis.

MASTER OF SCIENCE IN MATHEMATICS. Faculty of Sciences, University of Novi Sad, Serbia.

2019–2021

BACHELOR OF SCIENCE IN MATHEMATICS. Faculty of Sciences, University of Novi Sad, Serbia.

2016–2019

SKILLS

Programming & Scientific Computing Python, NumPy, SciPy, Pandas, scikit-learn, TensorFlow, Git, Linux, MATLAB, Mathematica.

Data Science Statistical Modeling, Predictive Analytics, Time Series Analysis, Machine Learning, Deep Learning, Clustering, Dimensionality Reduction, Data Visualization.

Mathematical & Quantitative Methods Dynamical Systems, Probability and Statistics, Network Science, Stochastic Processes, Mathematical Modeling, Numerical Analysis, Optimization, Scientific Computing.

Data Infrastructure & DevOps Docker, Kubernetes, Apache Superset, LakeFS, Invenio, Data Management.

Research & Professional Skills Independent Research, Complex Problem Solving, Algorithm Development, Quantitative Research, Interdisciplinary Research.

Communication & Collaboration

Technical Writing, Scientific Communication, Public Speaking, Science Outreach, Stakeholder Communication, Team Collaboration.

AWARDS & DISTINCTIONS

- ◇ Valedictorian (Class of 2019), Faculty of Sciences, University of Novi Sad.
- ◇ Valedictorian in Natural Sciences and Mathematics (Class of 2019), University of Novi Sad.
- ◇ Recipient of multiple national merit-based scholarships for academic excellence.